

34.2 Link functions

The linear model says $E(Y_i) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$. For a count that must be nonnegative, or a probability that must lie in $(0, 1)$, that equation is a promise the right-hand side cannot keep: a linear function of unbounded regressors takes every real value. The resolution is to put a monotone function between the mean and the linear predictor.

34.2.1 The model

Definition 34.2.1 — Generalized linear model

Let $\mathbf{X}$ be an $n \times p$ model matrix with rows $\mathbf{x}_{(i)}^\top$, and let $w_1, \dots, w_n > 0$ be known prior weights. A **generalized linear model** consists of

- (a) a **random component**: $Y_1, \dots, Y_n$ are independent, and Y_i has the density Eq. 34.1.1 with natural parameter θ_i , weight w_i and a dispersion ϕ that is the same for all i ;
- (b) a **linear predictor** $\eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, with $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta} \in \mathbb{R}^n$;
- (c) a **link function** $g : \mathcal{M} \rightarrow \mathbb{R}$, strictly monotone and twice continuously differentiable, with

$$g(\mu_i) = \eta_i, \quad \mu_i = E(Y_i) = b'(\theta_i).$$

The inverse $h = g^{-1}$ is the **response function**, so $\mu_i = h(\eta_i)$. The **admissible set** of the model is $\mathcal{B} = \{\boldsymbol{\beta} \in \mathbb{R}^p : \mathbf{x}_{(i)}^\top \boldsymbol{\beta} \in g(\mathcal{M}) \text{ for all } i\}$.

The linear model of [Chapter 5](#) is the case: normal random component, identity link, $w_i = 1$, $\phi = \sigma^2$, $\mathcal{B} = \mathbb{R}^p$. Every classical result of Parts II and III is about that one cell of a large table.

Three remarks. The dispersion is common to all observations: the regressors move the mean, the variance follows through V , and nothing else may vary (modelling ϕ too is the subject of [Chapter 45](#), Definition 45.5.1). The weights w_i are known, not estimated, and usually record group sizes. And g transforms the mean, not the data: [Chapter 22](#) transformed Y itself, whereas here the response is left alone and the link acts inside the expectation.

Definition 34.2.2 — Canonical link

The **canonical link** of a family is $g = (b')^{-1}$, so that $\theta_i = \eta_i$: the natural parameter *is* the linear predictor. Equivalently, $g'(\mu) = 1/V(\mu)$.

Family	canonical link	$g(\mu)$	range of η	other links in use
normal	identity	μ	$\mathbb{R}$	log, inverse
binomial	logit	$\log\{\mu/(1 - \mu)\}$	$\mathbb{R}$	probit, complementary log–log, identity
Poisson	log	$\log \mu$	$\mathbb{R}$	identity, square root

Family	canonical link	$g(\mu)$	range of η	other links in use
gamma	inverse	$-1/\mu$	$(-\infty, 0)$	log, identity
inverse Gaussian	inverse square	$-1/(2\mu^2)$	$(-\infty, 0)$	log
negative binomial	$\log\{\mu/(\mu + k)\}$	as shown	$(-\infty, 0)$	log

The sign conventions for the gamma and inverse Gaussian are cosmetic: $1/\mu$ and $-1/\mu$ give the same fitted values with $\boldsymbol{\beta}$ negated. The column headed “range of η ” is not. For the normal, binomial and Poisson the canonical link maps $\mathcal{M}$ onto all of $\mathbb{R}$, so $\mathcal{B} = \mathbb{R}^p$ and no constraint is ever active. For the gamma and inverse Gaussian it maps onto a half-line, and $\mathcal{B}$ is the polyhedron $\{\boldsymbol{\beta} : \mathbf{x}_{(i)}^\top \boldsymbol{\beta} < 0 \text{ for all } i\}$ — an open convex set with a boundary a fitting algorithm can walk into.

34.2.2 What a link choice costs and buys

Idea. The canonical link is a mathematical convenience, not a scientific claim. It makes the likelihood concave (Theorem 34.3.3), makes observed and expected information agree (Theorem 34.3.1), reduces the data to p sufficient statistics (Section 34.1), and forces the fitted values to reproduce the observed weighted column totals of $\mathbf{X}$ exactly (Corollary 34.3.2). None of that is a reason to believe the mean depends on the regressors in that particular way.

Four considerations decide a link in practice.

Range. A link whose range is all of $\mathbb{R}$ removes constraints from the fitting problem. The log link for a Poisson response guarantees $\hat{\mu}_i > 0$ for every $\boldsymbol{\beta} \in \mathbb{R}^p$, and the logit link for a binomial response guarantees $\hat{\mu}_i \in (0, 1)$. An identity link for either does not.

Example. (An identity link close to its boundary)

Eight counts were simulated from a Poisson model with mean $12 - 1.2x$ at $x = 1, \dots, 8$. An identity-link Poisson fit recovers the declining mean, and every fitted value is positive — but the fitted line reaches zero at $x = 9.89$, barely outside the range of the data. Predicting at $x = 10$ would return a negative mean, and a slightly different sample would push the boundary inside the observed range, where the likelihood is $-\infty$. The log link has no such boundary.

Interpretation. A log link makes the regressors act multiplicatively, $\mu = e^{\beta_0} \prod_j e^{\beta_j x_j}$, so β_j is a log rate ratio and e^{β_j} the factor multiplying the mean per unit of x_j . A logit link makes β_j a log odds ratio, the subject of Chapter 35 (Definition 35.1.1). An identity link keeps β_j a difference of means, often what a policy question asks about. These are different scientific statements, and the choice belongs on the scale where effects are plausibly additive, not on convenience.

Fit and stability. Different links give genuinely different mean curves, and a badly chosen one shows up in the residuals like any other misspecification. Non-canonical links can also produce a log-likelihood with several stationary points, and links with a restricted range need the fitting algorithm to be protected; Section 34.4 returns to this.

Example. (Food expenditure: three models, two decisions)

Engel's classic budget survey records annual income and food expenditure for 235 working-class households. Food expenditure is positive, right-skewed, and visibly more variable at high incomes. Write $u = \log(\text{income})$ and fit three models with the same linear predictor $\beta_0 + \beta_1 u$: a normal random component with the identity link (ordinary least squares), a gamma random component with the identity link, and a gamma random component with the log link. The first two differ only in the variance function, the last two only in the link.

The identity-link fits give $\hat{\beta}_1 = 571.02$ (normal) and 474.54 (gamma) francs per unit of log income. The gamma/log fit gives $\hat{\beta}_1 = 0.8629$, an elasticity: a one per cent rise in income is associated with a 0.86 per cent rise in expected food expenditure. Its estimated dispersion is $\hat{\phi} = 0.0177$, so by Example (The gamma and its constant coefficient of variation) the fitted coefficient of variation is $\sqrt{\hat{\phi}} = 0.133$, the same at every income.

Figure 34.2.1 separates the two decisions. In panel (a) the identity-link mean curve is concave in income and flattens, while the log-link curve is a power law that keeps climbing; at the largest income in the sample the two fitted means differ by a factor of 1.786. Panel (b) holds the link at the identity and changes the variance function, so its bands come from the identity-link fits, the gamma one using $\hat{\phi} = 0.0240$ and a coefficient of variation of 0.155. The normal band has constant width and at low incomes dips below zero; the gamma band widens in proportion to the mean, as the data do.

```
import numpy as np
import statsmodels.api as sm

engel = sm.datasets.engel.load_pandas().data
y = engel["foodexp"].to_numpy()                # food expenditure
X = np.column_stack([np.ones(len(y)), np.log(engel["income"].to_numpy())])

gauss = sm.GLM(y, X, family=sm.families.Gaussian()).fit()           # identity link
gam_id = sm.GLM(y, X, family=sm.families.Gamma(sm.families.links.Identity())).fit()
gam_log = sm.GLM(y, X, family=sm.families.Gamma(sm.families.links.Log())).fit()

for name, fit in [("normal/identity", gauss), ("gamma/identity", gam_id), ("gamma/
log", gam_log)]:
    print(f"{name:16s} coefficients {fit.params[0]:9.3f} {fit.params[1]:8.3f}"
          f"    dispersion {fit.scale:8.4f}")
```

[Run this code online](#)

34.2.3 Links and transformations are not the same thing

Transforming the response and choosing a link both replace μ by $g(\mu)$, but in different places. Modelling $E\{g(Y)\} = \mathbf{x}^\top \boldsymbol{\beta}$ changes the quantity modelled: by Jensen's inequality $E\{g(Y)\} \neq g\{E(Y)\}$ unless g is affine, so the fitted curve estimates the mean of a transformed variable, and returning to the mean of Y needs the retransformation correction of Proposition 22.5.1.

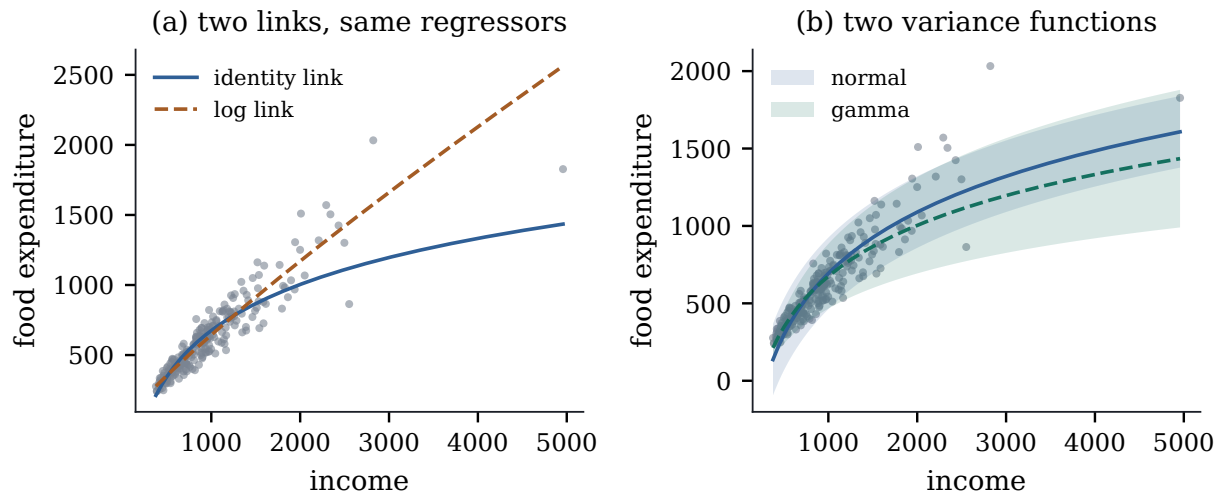


Figure 34.2.1: Engel's household budget data, $n = 235$. (a) Two links with the same gamma random component and the same regressors: the identity link (solid) and the log link (dashed). (b) Two variance functions with the same identity link: the normal band $\hat{\mu} \pm 2\hat{\sigma}$ has constant width and dips below zero, while the gamma band $\hat{\mu}(1 \pm 2\sqrt{\hat{\phi}})$, with $\hat{\phi}$ from the identity-link gamma fit, widens with the mean.

Modelling $g\{E(Y)\} = \mathbf{x}^\top \boldsymbol{\beta}$ leaves the response alone, and $\hat{\mu}_i = h(\mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}})$ estimates the mean of Y directly.

The transformation also had a second job the link does not do: stabilizing the variance (Theorem 22.4.1). A generalized linear model does not need that, because the variance function is already specified, and splitting one overloaded decision into two independent ones is the framework's main practical advantage. The weighted regression on empirical logits of Exercise (B2) is the historical compromise: the right link, but one weighted least squares step in place of the likelihood.

34.2.4 Exercises

34.2.4.1 A. Check your understanding

Exercise A1.

Describe the admissible set $\mathcal{B}$ of Definition 34.2.1 for (a) a Poisson model with the identity link, (b) a binomial model with the identity link, (c) a gamma model with the log link. Which of the three is all of $\mathbb{R}^p$?

34.2.4.2 B. Practice

Exercise B1.

Let Y be positive with $E(Y) = \mu$ and $\text{Var}(Y) = \phi\mu^2$, as in the gamma family. Using a second-order Taylor expansion of $\log Y$ about μ , show that $E(\log Y) \approx \log \mu - \phi/2$. Deduce that a least squares regression of $\log y_i$ on $\mathbf{X}$ and a gamma/log generalized linear model estimate intercepts that differ by approximately $\phi/2$, while their slopes agree to

this order. For the fit of [Example \(Food expenditure: three models, two decisions\)](#), how large is the discrepancy?

Solution. With $g(y) = \log y$, $g(Y) \approx g(\mu) + (Y - \mu)/\mu - (Y - \mu)^2/(2\mu^2)$; taking expectations gives $E(\log Y) \approx \log \mu - \text{Var}(Y)/(2\mu^2) = \log \mu - \phi/2$. Since the correction is the same constant for every observation, it is absorbed by the intercept when the model has one, and the slopes are unaffected to this order. For Engel's data $\hat{\phi} = 0.0177$, so the two intercepts differ by about $\hat{\phi}/2 = 0.0177/2 = 0.009$, less than one per cent on the expenditure scale. (For the gamma, $E(\log Y)$ can be computed exactly with the digamma function; the approximation is good when ϕ is small.)

Exercise B2.

A Poisson model is sometimes fitted with the square-root link $g(\mu) = \sqrt{\mu}$, because Theorem 22.4.1 makes $\sqrt{Y}$ approximately variance-stabilized. Find $\mathcal{B}$ for this link, find $d\mu/d\eta$, and explain why the link being variance-stabilizing for Y is irrelevant to whether it is a good link.

Solution. $g(\mathcal{M}) = (0, \infty)$, so $\mathcal{B} = \{\boldsymbol{\beta} : \mathbf{x}_{(i)}^\top \boldsymbol{\beta} > 0 \ \forall i\}$, and $\mu = \eta^2$ gives $d\mu/d\eta = 2\eta = 2\sqrt{\mu}$. Variance stabilization matters when the model is fitted by *least squares on the transformed data*, because least squares wants constant variance. A generalized linear model already knows $V(\mu) = \mu$ and weights each observation accordingly (Theorem 34.3.1), so the only question left about the link is whether $\sqrt{EY}$ is plausibly linear in the regressors.

Exercise B3.

The complementary log–log link for a binomial response is $g(\mu) = \log\{-\log(1 - \mu)\}$. Show that it is a bijection from $(0, 1)$ onto $\mathbb{R}$, compute $d\mu/d\eta$, and show that it is *not* symmetric: $g(\mu) \neq -g(1 - \mu)$, unlike the logit and probit links. Which tail does it approach more slowly?

Solution. $\mu \mapsto -\log(1 - \mu)$ maps $(0, 1)$ onto $(0, \infty)$ increasingly, and $\log$ maps $(0, \infty)$ onto $\mathbb{R}$, so g is an increasing bijection with inverse $h(\eta) = 1 - \exp(-e^\eta)$ and $d\mu/d\eta = e^\eta \exp(-e^\eta) = (1 - \mu)\{-\log(1 - \mu)\}$. Symmetry would require $h(-\eta) = 1 - h(\eta)$, which fails, for example at $\eta = 0$, where $h(0) = 1 - e^{-1} = 0.632 \neq \frac{1}{2}$. As $\eta \rightarrow -\infty$, $\mu \approx e^\eta \rightarrow 0$ exponentially, while as $\eta \rightarrow +\infty$, $1 - \mu = \exp(-e^\eta) \rightarrow 0$ doubly exponentially: the approach to 1 is far faster than the approach to 0. [Chapter 35](#) derives the link from a proportional-hazards argument (Proposition 35.2.1).

34.2.4.3 C. Going deeper

Exercise C1.

Every link in the table except the two for a binomial response is a power of μ or a logarithm. That is not an accident. For a positive response call a link g on $(0, \infty)$ **scale-equivariant** if for every $c > 0$ there are constants $a(c) \neq 0$ and $b(c)$ with $g(c\mu) = a(c)g(\mu) + b(c)$ for all $\mu > 0$.

- (a) Check that $g(\mu) = \mu^\lambda$ with $\lambda \neq 0$ and $g(\mu) = \log \mu$ are scale-equivariant, and identify $a(c)$ and $b(c)$.
- (b) Conversely, let g be strictly monotone and twice continuously differentiable on $(0, \infty)$, as [Definition 34.2.1](#) requires, and scale-equivariant. Show that $g(\mu) = A\mu^\lambda + B$ for some $A \neq 0$, $\lambda \neq 0$ and B , or $g(\mu) = A \log \mu + B$ with $A \neq 0$. (Put $G(t) = g(e^t)$, differentiate the defining relation in t , and solve the resulting equation for G' .)
- (c) Explain what this means for a model with an intercept: if the response is measured in other units, $Y \mapsto cY$, the class of fitted mean curves $x \mapsto h(\mathbf{x}^\top \boldsymbol{\beta})$ is unchanged exactly when the link is scale-equivariant, the coefficients being relabelled as $\boldsymbol{\beta} \mapsto a(c)\boldsymbol{\beta} + b(c)\mathbf{e}_1$.
- (d) Show that $g(\mu) = \mu - 1/\mu$, a perfectly legal increasing bijection from $(0, \infty)$ onto $\mathbb{R}$, is not scale-equivariant, so that the model it defines is a different model for expenditure in francs and for expenditure in centimes.

Solution.

- (a) $(c\mu)^\lambda = c^\lambda \mu^\lambda$, so $a(c) = c^\lambda$ and $b(c) = 0$; and $\log(c\mu) = \log \mu + \log c$, so $a(c) = 1$ and $b(c) = \log c$.
- (b) With $G(t) = g(e^t)$ the relation reads $G(t+s) = a(e^s)G(t) + b(e^s)$ for all real t, s . Differentiating in t gives $G'(t+s) = a(e^s)G'(t)$; setting $t = 0$ gives $G'(s) = a(e^s)G'(0)$. If $G'(0) = 0$ then $G' \equiv 0$ and g is constant, contradicting strict monotonicity, so $G'(0) \neq 0$ and $\psi = G'/G'(0)$ satisfies $\psi(t+s) = \psi(t)\psi(s)$, $\psi(0) = 1$. Since G is twice continuously differentiable, ψ is differentiable; differentiating in s at $s = 0$ gives $\psi'(t) = \lambda\psi(t)$ with $\lambda = \psi'(0)$, so $\psi(t) = e^{\lambda t}$ and $G'(t) = G'(0)e^{\lambda t}$. Integrating, $G(t) = G(0) + G'(0)(e^{\lambda t} - 1)/\lambda$ if $\lambda \neq 0$, and $G(t) = G(0) + G'(0)t$ if $\lambda = 0$. Writing $\mu = e^t$ gives $g(\mu) = A\mu^\lambda + B$ with $A = G'(0)/\lambda$, or $g(\mu) = A \log \mu + B$ with $A = G'(0)$.
- (c) If $g(\mu_i) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ for all i , then $g(c\mu_i) = a(c)\mathbf{x}_{(i)}^\top \boldsymbol{\beta} + b(c)$, which is $\mathbf{x}_{(i)}^\top \{a(c)\boldsymbol{\beta} + b(c)\mathbf{e}_1\}$ when the first column of $\mathbf{X}$ is $\mathbf{1}$. So the rescaled means satisfy the same model with relabelled coefficients, and the set of mean curves the model can produce is the same set. If g is not scale-equivariant, the two sets differ, and which model fits better depends on the units — a property no scientific model should have.
- (d) Suppose $c\mu - 1/(c\mu) = a(\mu - 1/\mu) + b$ for all $\mu > 0$. The three functions μ , $1/\mu$ and 1 are linearly independent on $(0, \infty)$, so $a = c$, $a = 1/c$ and $b = 0$, forcing $c^2 = 1$. Only $c = 1$ works. By (b) this had to happen: $\mu - 1/\mu$ is a sum of two powers and not of the permitted form.

The conclusion also explains why the binomial links escape: a probability is a pure number, so there are no units to change, and the logit and complementary log-log are free to be what the science suggests.

Exercise C2.

[Exercise \(B1\)](#) compares “transform the data” with “transform the mean” to first order in ϕ . For some families the comparison is exact.

- (a) Suppose $Y = \mu W$ where the distribution of W does not depend on μ , $\mathbb{E} W = 1$ and $\mathbb{E} |\log W| < \infty$ — a **scale family**. Show that $\mathbb{E}(\log Y) = \log \mu + \kappa$ with

$\kappa = E \log W$, a constant, and that $\kappa < 0$ unless $W \equiv 1$.

- (b) Deduce that if $\log \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ and $\mathbf{X}$ has an intercept, the population least squares regression of $\log Y$ on $\mathbf{X}$ has coefficients $\boldsymbol{\beta}$ with the intercept lowered by exactly $|\kappa|$: the slopes agree *exactly*, not merely to first order.
- (c) Show that the gamma of Example (The gamma and its constant coefficient of variation) with shape $\nu = 1/\phi$ is a scale family, and that $\kappa = \psi(\nu) - \log \nu$, where $\psi = \Gamma'/\Gamma$ is the digamma function. (Differentiate $\int_0^\infty g^{\nu-1} e^{-g} dg = \Gamma(\nu)$ with respect to ν .) The standard expansion $\psi(\nu) = \log \nu - 1/(2\nu) - 1/(12\nu^2) + \dots$ then returns $\kappa \approx -\phi/2$.
- (d) Show that no such exact statement is available for a Poisson response. Show that the Poisson is not a scale family, that $E(\log Y) = -\infty$ whenever $\Pr(Y = 0) > 0$, and that even the patched $\log(1 + Y)$ has $E \log(1 + Y) - \log \mu \rightarrow \infty$ as $\mu \rightarrow 0$ and $E \log(1 + Y) - \log \mu \leq \log(1 + 1/\mu) \rightarrow 0$ as $\mu \rightarrow \infty$, so the gap is far from constant.

Solution.

- (a) $\log Y = \log \mu + \log W$ and $\log W$ has a law free of μ , so $E \log Y = \log \mu + \kappa$ with $\kappa = E \log W$. By Jensen's inequality for the strictly concave $\log$, $\kappa = E \log W < \log E W = 0$ unless W is degenerate, in which case $W \equiv 1$ and $\kappa = 0$.
- (b) The vector with entries $E \log Y_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \kappa$ is $\mathbf{X} \boldsymbol{\beta}^*$ with $\boldsymbol{\beta}^* = \boldsymbol{\beta} + \kappa \mathbf{e}_1$, because the first column is $\mathbf{1}$. The population least squares coefficients are those of the best linear predictor of $\log Y$, which here reproduces the mean exactly; by Proposition 5.5.7 they are $\boldsymbol{\beta}^*$. Every coefficient but the intercept is β_j itself, and $\kappa < 0$.
- (c) A gamma response with mean μ and shape ν is $Y = (\mu/\nu)G$ with $G \sim \text{Gamma}(\nu, 1)$, so $W = G/\nu$ has $E W = 1$ and a law free of μ . Differentiating $\int_0^\infty g^{\nu-1} e^{-g} dg = \Gamma(\nu)$ under the integral sign gives $\int_0^\infty (\log g) g^{\nu-1} e^{-g} dg = \Gamma'(\nu)$, that is $E \log G = \psi(\nu)$. Hence $\kappa = E \log G - \log \nu = \psi(\nu) - \log \nu$, which is negative, as (a) requires. With $\nu = 1/\phi$ the expansion gives $\kappa = -\phi/2 - \phi^2/12 - \dots$, the leading term of [Exercise \(B1\)](#), and the next term is smaller than it by the factor $6/\phi$, which for Engel's fit is several hundred.
- (d) For a scale family $\text{Var}(Y)/(E Y)^2 = \text{Var}(W)$ does not depend on μ ; for the Poisson it is $\mu/\mu^2 = 1/\mu$, so the Poisson is not one. Since $\Pr(Y = 0) = e^{-\mu} > 0$ and $\log 0 = -\infty$, $E \log Y = -\infty$ and the transformed regression is not even defined. For the patch, on the one hand $0 \leq E \log(1 + Y) \leq E Y = \mu \rightarrow 0$ as $\mu \rightarrow 0$ by $\log(1 + y) \leq y$, while $\log \mu \rightarrow -\infty$, so the gap diverges; on the other hand Jensen's inequality gives $E \log(1 + Y) \leq \log(1 + \mu)$, so the gap is at most $\log(1 + 1/\mu)$, which tends to 0. A gap that ranges over the whole positive half-line cannot be absorbed into an intercept, so the two regressions have genuinely different slopes, and only the link version answers a question about $E(Y)$.